

50. THE CRANIUM

DURATION : 01:00:35

STUDY SHEET

COURSE SUMMARY

This video explores in depth the influence of brain development on the intracranial membranes and the venous system, focusing on the formation of the reciprocal tension membranes. You will learn how the base of the skull is crucial for the cranosacral mechanism, and how the teeth and tongue act as postural sensors that influence balance and respiration. The interaction between the inner ear, posture, and bodily rhythms is also discussed to provide a holistic view of neuro-digestive balance. Finally, the video addresses the metabolic processes related to cartilaginous formation and tissue chains, illustrating their significance in overall bodily dynamics.

THE CRANIUM

THE INFLUENCE OF CEREBRAL DEVELOPMENT ON INTRACRANIAL MEMBRANES AND THE VENOUS SYSTEM

Cerebral development guides and organises the phenomena of the **intracranial membranes**. This process is essential for the formation of the **reciprocal tension membrane** system.

This system is crucial for the pathway of **vessels**, particularly important veins, serving to drain and release **CO2**. Many problems arise from **stases**, with key points such as the **pterions** or certain **transverse occipital venous sinuses**, where eight layers meet. This is a vital crossroads.

THE CRANIO-SACRAL MECHANISM AND THE TEETH

The **base of the cranium** is a fundamental reference point of the **cranio-sacral** mechanism. The **teeth** play an essential role as postural sensors, especially when the child straightens up and begins to have teeth.

They provide support for **correct breathing**. The absence of teeth in the elderly, for example, alters posture and breathing, illustrating their importance.

ROLE OF TEETH AND TONGUE IN POSTURE AND BALANCE

Teeth are sensors that contribute to the child's straightening and improved posture. Not missing a tooth is crucial, as its absence can shift the tongue to the side.

The **tongue** is a key element of the **lingual axis**, which is part of the longitudinal axis and the embryonic chain extending to the thumbs and big toes. The tongue, posture, and teeth work together to maintain **balance**.

THE INNER EAR, POSTURE, AND BODY RHYTHMS

Stability is sought via the **vestibular system**, with an orientation of 23 degrees for the inner ear. This angulation horizontalises and aligns with that of the heart and head tilt.

The body orientates itself in relation to the universe on angular planes to regain stability. We have **cardiac rhythms** (100,000 beats/day), **respiratory rhythms** (25,000 breaths/day), and **digestive rhythms** (1 to 2 litres swallowed/day).

THE BASE OF THE CRANIUM, THE PITUITARY GLAND, AND NEURO-DIGESTIVE BALANCE

The base of the cranium is a major articulation between the brain and the aorta. A crucial **embryonic flexion**, resulting from the bending of the embryo and the encephalon, compresses the mesenchyme under the mesencephalon, forming the **sub-mesencephalic septum**.

The heart, aortas, and amniotic cavity organise this space. A small endodermal pouch, **Rathke's pouch**, forms the anterior pituitary, composed of a neurological part and a digestive part. The **infundibulum**, coming from the brain, joins it to form the neurological pituitary.

The pituitary gland, the endocrine and neurological conductor, rests on the base of the cranium, a point of support. This base of the cranium is a balance between the **neurocranium** and the **viscerocranium**, reflecting the environment of the brain and the digestive system.

METABOLIC PROCESSES AND CARTILAGINOUS FORMATION

Torsion phenomena or imbalances in the movement of the base of the cranium can affect pituitary function. Function is a repetition of a primary embryological movement that organises **motility** and **motoricity**.

The formation of Rathke's pouch is also a field of **metabolic contusion**, where two tissues contract. Initially, cartilaginous tissue develops the hypophyseal, parachordal, and prechordal cartilages, which will form the hypoxephital base, the **sphenoid**, the **ethmoid**, and the **petrous pyramids**.

TISSUE CHAINS AND THEIR SIGNIFICANCE

The **occipito-lingual chain** is crucial for the complete dynamics of the body and constitutes a midline. The lingual complex and the tongue (heart bud) are linked to the **thyroid**, which electrifies our words.

The **frontal sinuses** are involved in self-expression and self-protection. **Mantras** are frequencies that neutralise incessant inner discourse, bringing us back to archetypal sounds.

NEUROLOGICAL CHAINS AND THEIR ARCHETYPAL ROLE

- **The lingual chain:** Interacts with the occipito-lingual chain.
- **The sphenoidal and axial chain:** A descending chain of life anticipation.

DEVELOPMENT OF THE CRANIUM AND BRAIN

The **neurocranium** and **viscerocranium** are separated by the base of the cranium, a vital element for development and articulation between the aorta and the brain. This is a precursor image of the cranium, with **cerebralisation** as the driving force.

The differential growth of the brain leads to the flexion of the embryo, with the **prosencephalon**, **mesencephalon**, and **rhombencephalon** providing the dynamics.

EMBRYONIC FLEXION AND THE BASE OF THE CRANIUM

The forward flexion of the encephalon makes the mesenchyme under the mesencephalon perpendicular to the longitudinal axis. This sub-mesencephalic compression forms the sub-mesencephalic septum, which is the origin of the base of the cranium.

Rathke's pouch, of endodermal origin (intestine), becomes the anterior pituitary, showing that the base of the cranium is aware of the environment of the brain and the digestive system.

THE MEMBRANES OF THE BRAIN: FROM DURA MATER TO BONE

The **neural tube** is surrounded by an **ectomeninge**, the anlage of the dura mater. This tissue develops along the axis of the neural tube, closing at the anterior and posterior neuropores.

With **telencephalisation**, the meninges grow and the midline approaches to form the **crista galli**. The tentorium cerebelli folds at the back, and the falx cerebri forms at the top, following the development of the brain.

DURAL BELTS AND CRANIAL OSSIFICATION

The dura mater, with its double layer (visceral and parietal), is an envelope of the brain. Areas of **densification**, called **dural belts** (anterior, middle, posterior), orient towards the base of the cranium, a place of intense membranous tension.

These mesodermal condensations, under the influence of metabolic growth fields, transform into **bone**, forming the cranial vault.

THE PROCESS OF TELENCEPHALISATION AND THE CRISTA GALLI

Initially, the anterior and posterior brain meet and compress the base of the cranium. The future **interorbital ligament** is present. During telencephalisation, the system is pulled upwards, straightening the **pterygoids** and developing the falx cerebri.

The crista galli, a crucial place for body and mind, is formed by the folding of tissue and the approximation of the pterygoids. It is a point of support for cerebralisation, the face, and the orientation of the eyes, leading to **clairvoyance**.

HEMISPHERIC SPECIALISATION AND ITS IMPLICATIONS

The embryonic flexion force transforms into a straightening force, developing the **cerebral hemispheres**. These attach to the crista galli, tightening up to the Egnon.

Each step, from **Nazion** to **Asterion**, releases movements up to the cranium. Good **dental occlusion** is essential for the freedom of cranial movements, especially in the frontal areas.

BONE FORMATION AND THE ROLE OF THE MESODERM

Bone is built within the membrane, and the brain is the engine of this development. The mesodermal tissue around the neural tube densifies to form the visceral dura mater, then the parietal, and finally the bone.

This mesoderm undergoes metabolic fields of growth and expansion, hardening to form the vault. Bone is the result and the finality of the **cerebralisation** process.

THE TAI CHI OF THE BRAIN AND THE TONGUE

The **ascensus** process of the brain, with its rapid growth, brings structures closer to the midline and spreads them laterally. The primitive digestive tube rises, and the tongue follows its development.

The depth of the tongue is a reflection of the depth of cerebralisation, a homologous symmetry.

THE BASE OF THE CRANIUM: A POINT OF EMBRYOLOGICAL CONVERGENCE

The base of the cranium is a point of convergence of information:

- **Bony:** hypophyseal and parachordal cartilages.
- **Endodermal and ectodermal.**
- **Brain tension membranes.**
- **Lines related to the branchial arches.**

It is a **hypothalamic-pituitary** convergence point, exchanging chemical and tonic information. The vascular roof of the pituitary is filled with vessels.

DURAL SHEATHS AND CRANIAL CONSTRUCTION

The primitive **dural sheaths** organise along the axis between the aorta and the brain, a major confluence. We also find the falx cerebri, the dural sheaths, and the bone that is built from a common tissue.

Cephalisation, the growth of the brain, organises this entire process, including the ventricular system and the tension membranes.

TISSUE CONTINUITY AND RADICULAR PROBLEMS

The external and internal dura mater, the **arachnoid**, and the **pia mater** constitute a continuity with the epineurium, perineurium, and endoneurium. Small mesodermal blood capillaries in the endoneurium are causes of stases.

Pumping and releases decongest radicular problems. **Hoffman's ligaments** and **Forestier's opercula** create valves for fluid balance, protecting nerve roots.

CRANIAL OSSIFICATION AND DURAL WINDOWS

The cranial vault is the result of densification. Metabolic fields participate in the **synchondrosis** of the base of the cranium. Detraction forms the bones in the direction of the displacements of the **trabeculae** and **osteoblasts**.

Ossification nuclei form at the periphery. The development of the brain, framed by the dural sheaths (like windows), dictates the chronology of ossification: frontal, parietal, superior occipital, and superior temporal.

The frontal, parietal, and occipital windows are the future sutures.

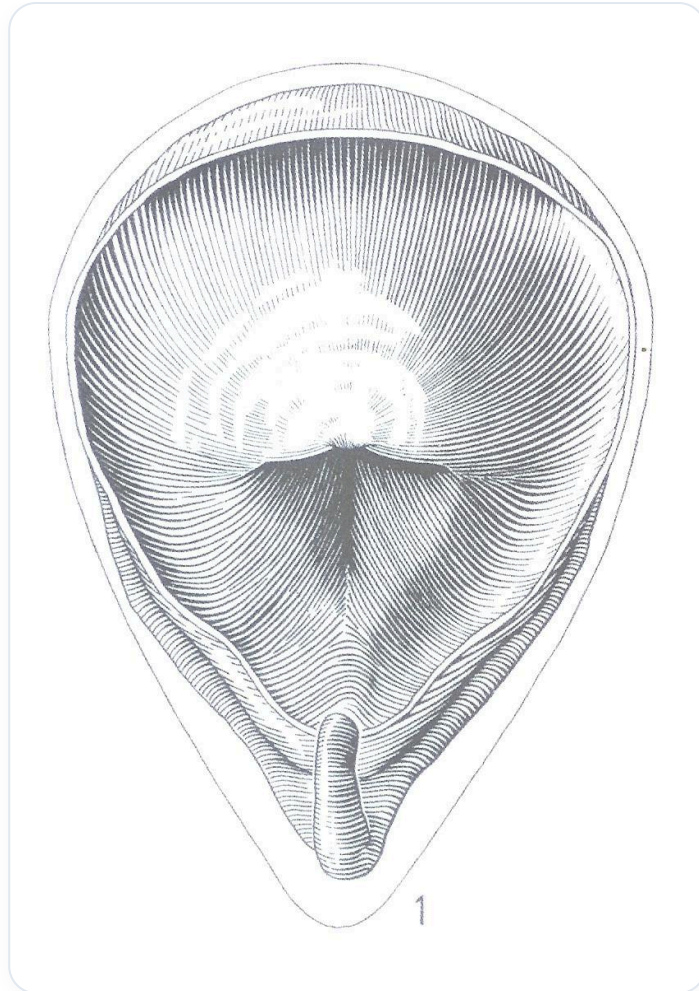


FIG. — *Urinary bladder*